

PlotTrajectory Instructions

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This is the **third step** of chemotaxis analysis in I2Tumbles. PlotTrajectory.m is used for either looking up a single bacterium for deeper analysis or quickly validate the results of CellTrackAnalysis.m by showing the tracking results of 20 random bacteria, using data from SimpleTrackingoutput.mat and pre-computed variables from CellTrackAnalysis.m.

Step 1: PlotTrajectory Parameters

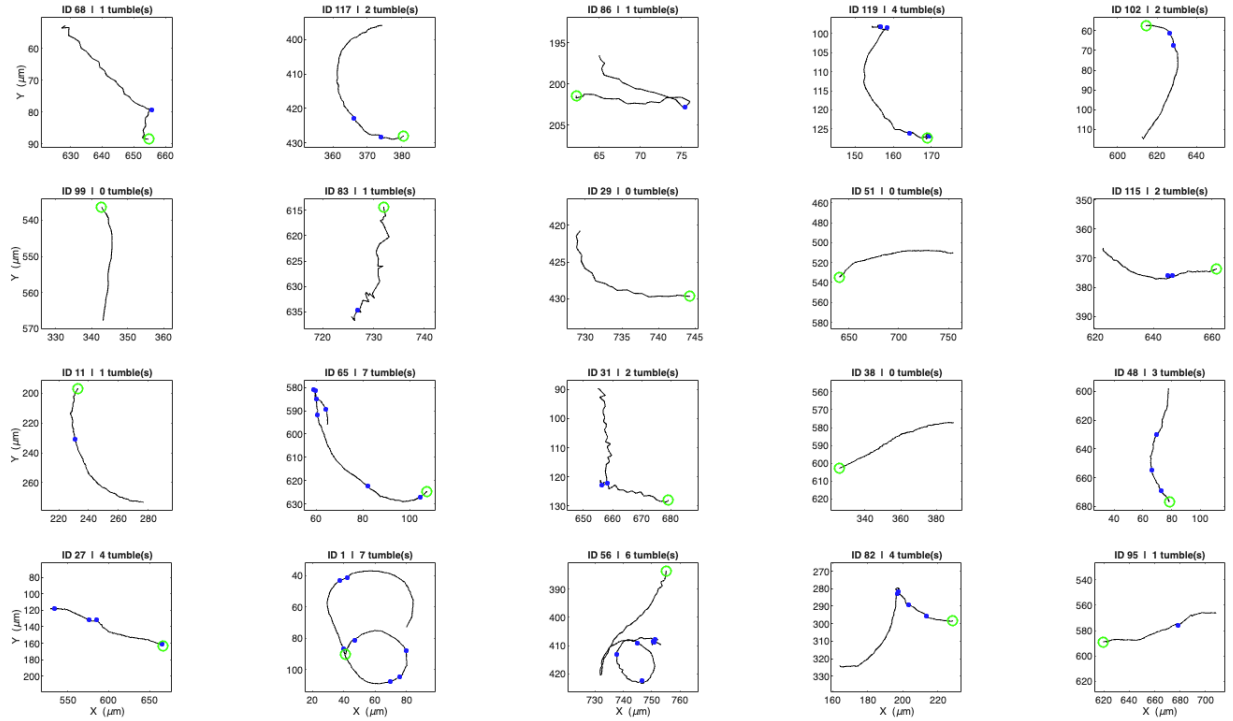
1. Open PlotTrajectory.m
2. Adjust parameters.

Parameter Name	Value Range	Data Type	Description
bacteria_idx	>= 0	Integer	When bacteria_idx > 0: The ID of the bacterium you want to look up. You can use TrackID column from CellTrackAnalysis.xlsx, or the ID number displayed in the scatterplot after running CellTrackAnalysis.m. When bacteria_idx = 0: Shows random 20 bacteria tracks.
secondVideo	> 0	Integer	Duration of the video in second.
raw_mat		String	Change the .mat file name if necessary

Step 2: Interpret Results

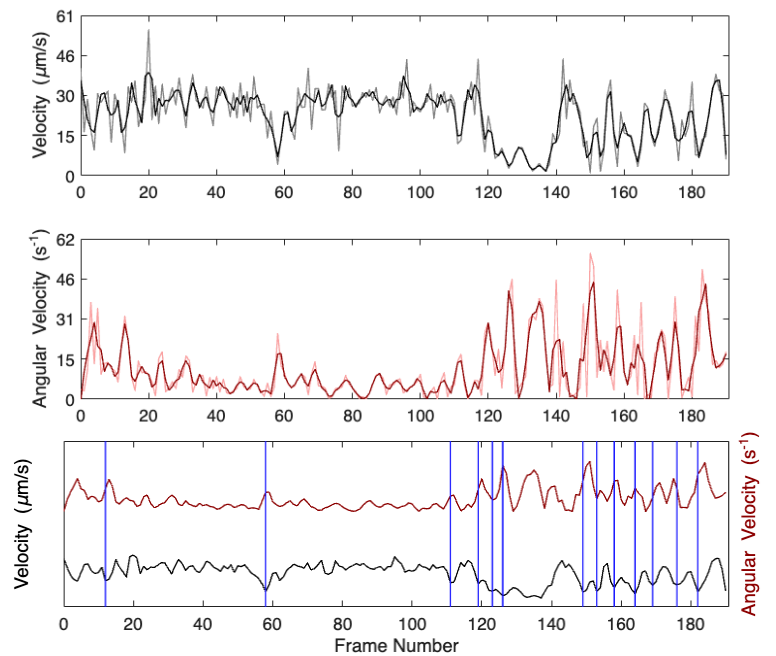
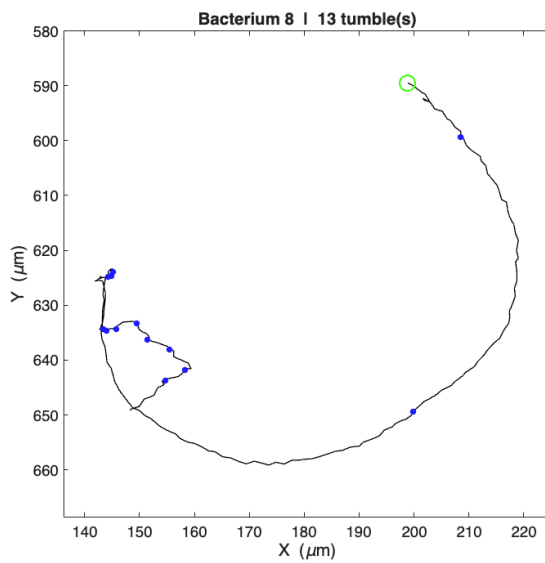
1. When bacteria_idx = 0
 - a. This is a quick way to see if the parameters set in CellTrackAnalysis.m works well for your data. This will show the tracks with tumbles (blue dots) of 20 random bacteria.

Random Sample of 20 Trajectories



2. When `bacteria_idx > 0`

- Input a specific bacterium's ID and see if the tumble calculation is accurate. Great for looking at outliers or specific data points.
- The algorithm should accurately define where the local minima of linear velocity and the local maxima of angular velocity meet as tumble points. See the blue vertical lines below.



Step 3: Go back to `CellTrackAnalysis.m` and tune the parameters if necessary

- If the parameter setting is too sensitive (catching too many tumbles), increase `depth_speed_ratio` or set to `fps_scale_sg_window=false`; and increase `sg_window` by 2.
- Do the opposite if you'd like to see more tumbles.

References

- Johnson et al. Run-and-tumble kinematics of *Enterobacter* Sp. SM3. *Physical Review*. E vol. 109,6–1 (2024). [[link](#)]
- Cell Track Analysis Algorithm GitHub: <https://github.com/Tang-BioPhys/SM3-Motility/tree/main>